

When does cortical coherence carry shared content?

Compatibility and self-reconstruction as testable conditions

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Abstract

Synchrony accounts treat collective phase order as the mechanism that integrates distributed cortical activity. We show formally that shared content and a self-model each need a condition beyond phase order, and state it. Three conditions are separated: coherence (collective phase order), compatibility (agreement of independently decoded local descriptions where regions overlap) and self-reconstruction (accurate recovery of the global state from an internal encoding). Compatible local descriptions determine a unique global state. Coherence bounds disagreement between overlapping descriptions only through shared encoders of bounded sensitivity, and the bound grows as the square root of the population read: on a simulated cortical sheet it constrains at most 10 sites, chaining through overlapping patches extends it to 0.312 mm, and across distant regions the link from coherence to content is an assumption. A contracting self-map fixes one state but reconstructs accurately only states within a diameter set by its error and contraction factor. Lean 4 verifies these implications; one model

satisfies all connecting hypotheses at once, showing consistency, not cortical realization. The conditions yield a disconfirmable comparison: at matched phase coherence and single-region decoding accuracy, decoders fitted independently on two regions should agree more, beyond their separate accuracies, on perceived than on unperceived stimuli classified by a no-report marker. Choosing the overlapping regions by what each decodes makes the unity correlate operational relative to a pre-registered decoder and criterion. An exploratory EEG analysis is a measurement check only. Unity and a minimal self remain interpretations that need an independent experiential endpoint.

1 Introduction

Endogenous extracellular fields can affect spike timing (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015) and may coordinate activity across cortical regions (McFadden, 2002, 2020; Pockett, 2012). Coordination raises a further question: when do those regions describe one situation, and when does the resulting state represent its own place in it?

Consider reaching for a cup. Vision and movement can describe different features while agreeing on the cup’s location. Matching their timing leaves agreement about that shared quantity untested. An internal description of the perceiver’s relation to the cup asks for still more. We call the three requirements *coherence* (collective phase order), *compatibility* (agreement of local descriptions on overlaps), and *self-representation* (accurate reconstruction from an internal encoding). The mathematical states are spatial measures and fields, compared with perceptual content through independently fitted decoders (Section 10.2).

16 Binding-by-synchrony and communication-through-coherence accounts
 17 treat phase relations as the medium of integration or of selective routing
 18 (Singer & Gray, 1995; Fries, 2015), and field theories place integration in the
 19 endogenous electromagnetic field (McFadden, 2020; Pockett, 2012). Both
 20 leave open what fact about synchronized activity makes it one content rather
 21 than coordinated activity carrying several. The contribution here is to sep-
 22 arate the three requirements and prove which follow from which; each result
 23 names a measurement that phase data alone cannot supply.

24 The objection that synchrony does not fix content is older than this work.
 25 Critiques of the temporal binding hypothesis argued that synchrony, where
 26 it occurs, cannot say which features belong together or what they jointly
 27 represent: that grouping has to be computed before synchrony could sig-
 28 nal it (Shadlen & Movshon, 1999). The formal results here are elementary:
 29 uniqueness of gluing, Banach’s fixed-point theorem, the triangle inequality
 30 and a pigeonhole count. What formalization adds to the informal argument
 31 is quantitative and bookkeeping content. It gives the rate at which a co-
 32 herence bound on content weakens with population size, the population at
 33 which it stops constraining anything and how far chaining through overlap-
 34 ping patches extends it, the diameter within which one contracting map can
 35 reconstruct, and the number of codes a separated family of states requires.
 36 It also gives a complete inventory of the premises a synchrony-to-content
 37 argument consumes, each of which names a measurement, together with one
 38 model in which all of them hold at once.

39 The familiar noisy mean-field oscillator supplies a stationary coherence
 40 threshold (Sakaguchi, 1988; Strogatz & Mirollo, 1991); we add a minimum
 41 cost for changing coherence quickly. A Lean 4 development checks the im-
 42 plications, and a joint witness shows that the eight connecting hypotheses of
 43 the formal composition are jointly satisfiable.

44 The empirical question is which measurements could support or discon-
 45 firm the proposed connections. A phase resultant can miss a locally ordered
 46 winding field; stationary phase distributions can leave the coupling mecha-
 47 nism unidentified; and coherent adaptation can lose alignment with a desig-
 48 nated content template. We therefore distinguish stationary branches from
 49 finite observed onset, and compare prespecified field and connectivity onset
 50 models on held-out emergence recordings. The comparison tests a specified
 51 spatial prediction, not the presence of experience. Section 10.2 states the
 52 comparison that bears on conscious content directly.

53 The article makes six claims:

- 54 1. Compatible local descriptions determine a unique global state, while
 55 equal phases coexist with incompatible descriptions (Section 4).
- 56 2. Coherence bounds disagreement between overlapping descriptions
 57 through shared encoders of bounded sensitivity, by $L\sqrt{2N(1-r^2)}$
 58 on the population read; on a simulated cortical sheet this constrains
 59 at most 10 sites, and chaining through overlapping patches extends it
 60 to 0.312 mm (Section 4.2).

- 61 3. A contracting self-map fixes one state and reconstructs within error ε
62 only a family of diameter at most $2\varepsilon/(1 - \Lambda)$; reconstructing M states
63 pairwise more than 2ε apart takes M distinguishable codes (Section 5).
- 64 4. Changing coherence in time τ has a minimum current cost that grows
65 as $1/\tau$ (Section 3.2).
- 66 5. Stationary phase statistics leave the coupling mechanism unidentified,
67 finite ramps shift the observed onset, and a winding field keeps local
68 order at global $r = 0$ (Section 8).
- 69 6. The link to content predicts that, at matched phase coherence and
70 single-region accuracy, decoders fitted independently on two regions
71 agree more, beyond their separate accuracies, on perceived than on
72 unperceived stimuli; equal excess counts against the link to content
73 (Section 10.2).

74 Sections 2–5 develop the three requirements; Section 6 applies their mea-
75 surement obligations to a digital candidate. Section 7 states what the formal
76 composition establishes, and Sections 8–9 examine observables and the corti-
77 cal test, and Section 11 states the scope of every result in one place. Table 1
78 lists the connecting hypotheses the article argues from; Supplemental Mate-
79 rial gives the full chain, derivations and controls. For a theoretical article the
80 Methods are the definitions, proofs and simulations. Each result is stated in
81 the section that uses it, Supplemental Material Table S1 maps it to its Lean
82 declaration, and every simulation’s model, parameters and controls are given

83 in Supplemental Material. The Results are the theorems of Sections 3–5 and
the numerical controls of Section 8.

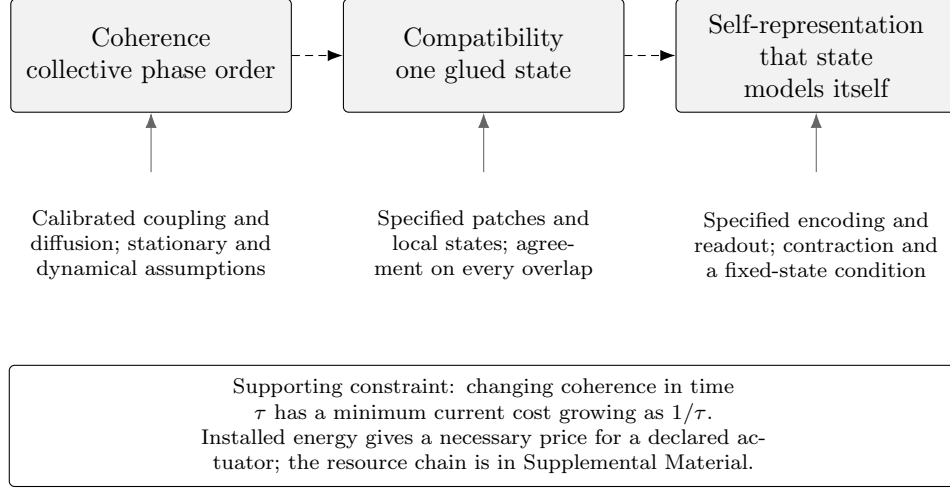


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections, each of which needs the inputs of the condition it points to in addition to phase coherence. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the connecting hypotheses of the formal composition that the article argues from.

84

85 2 Limits of phase order

86 Several limits hold without the cortical hypothesis. Equal phases can coexist
87 with incompatible local descriptions; phase order constrains shared content
88 only through encoders related on overlaps. A winding field can have zero
89 global resultant while remaining locally ordered. A constant readout can have
90 a unique fixed point without reconstructing its input. A region encodes only

the states it distinguishes, and by a declared deadline a site can report only changes inside its causal past. These limits reflect familiar phase oscillator and information constraints (Acebrón et al., 2005; Ott & Antonsen, 2008; Dörfler & Bullo, 2014; Breakspear, 2017). The sections below state each one beside the condition or measurement it motivates.

3 Coherence: physical coupling and its cost

3.1 Dynamics and the stationary threshold

The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here θ_i is the phase of oscillator i , ω_i its natural frequency, $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized

stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

where I_n are modified Bessel functions and ψ is the collective phase. The formal derivation uses the absence of stationary current for a periodic gradient drift to obtain this density, then verifies normalization and self-consistency. At $K \leq K_c = 2D$, only $r = 0$ is self-consistent (Sakaguchi, 1988; Strogatz & Mirollo, 1991). Above threshold, exactly one positive solution coexists with zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \rightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero. These linear estimates concern the limiting equation. Relating them to finite stochastic trajectories requires a dynamical mean-field limit and branch selection; exchange symmetry alone does not establish the needed independence (Sznitman, 1991).

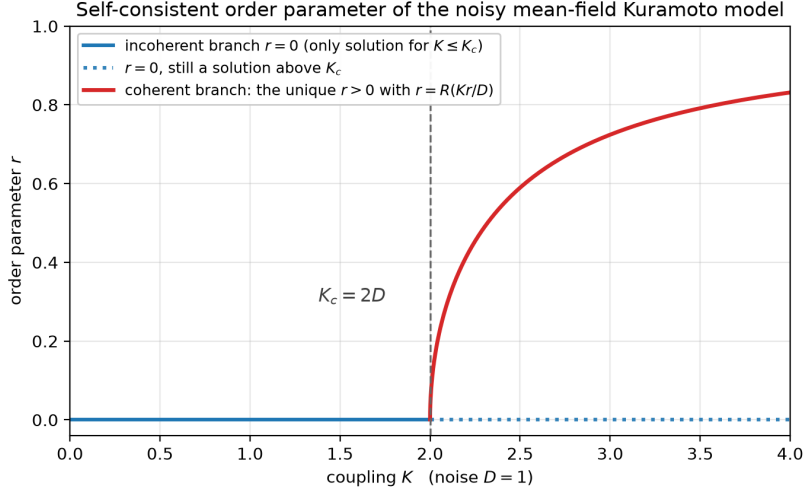


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

123 3.2 A speed limit on changes of coherence

124 One thermodynamic statement bears on coherence directly. For a smooth
 125 positive periodic phase density with probability current J and diffusion D ,
 126 the current functional $\sigma_J = \int J^2 / (D\rho) d\theta$ satisfies

$$\int_0^\tau \sigma_J dt \geq \frac{[\arcsin r(\tau) - \arcsin r(0)]^2}{D\tau} \quad (4)$$

127 while $0 < r < 1$. A change of coherence completed in time τ therefore has a
 128 minimum current cost that grows as $1/\tau$, whereas holding a stationary order
 129 has no positive floor. Read the other way, a current budget C sets a minimum

130 duration $\tau \geq [\Delta \arcsin r]^2 / (DC)$ for a given change. This bears on ignition-
 131 like transitions, in which large-scale coherence rises abruptly (Dehaene et al.,
 132 2014), and on recovery latency after anaesthesia: each has a floor set by the
 133 current available to drive it. The bound is a heat or power statement only
 134 when J is identified with a physical flux, a reservoir at a stated temperature
 135 is specified and local detailed balance holds for the steps concerned; only
 136 then does σ_J convert to entropy production (Landauer, 1961; Norton, 2005;
 137 Shenker, 2000). Installed energy supplies a separate, necessary price: for
 138 a declared actuator whose response modes carry installation prices, mean
 139 coupling is at most κU_{inst} for a mode-data factor κ , so a coherent stationary
 140 branch requires installed energy $U_{\text{inst}} > 2D/\kappa$. Supplemental Material gives
 141 both derivations, the register and feedback budgets, and the sensitivity of
 142 the metabolic comparison to its assumptions.

143 **4 Compatibility: from coordinated activity** 144 **to a global description**

145 **4.1 Why common timing is insufficient**

146 The formal setting is a sheaf: each patch of the substrate carries local de-
 147 scriptions, and restriction maps cut a description on a patch down to any
 148 smaller patch. A *section* is a description on one patch, and a *global section* is
 149 a description on the whole substrate. For overlapping patches U and V , com-

150 patibility means that restricting either local description to $U \cap V$ gives the
151 same object. Descriptions can agree on shared quantities while contributing
152 different information elsewhere. Compatible sections of the specified sheaf
153 have a unique global extension. On a finite cover, local finite spatial mea-
154 sures agreeing as measures on every overlap also have a unique finite-measure
155 extension whose sheaf image is the glued section. No mass-one normalization
156 is imposed.

157 In the cup example, visual and reaching descriptions must express a
158 shared quantity, such as cup location, in common coordinates. Restriction
159 retains that quantity while dropping details particular to each description.
160 Agreement permits surface appearance and hand configuration to contribute
161 to one situation. The mathematical counterexample assigns the same phase
162 to every patch while two local mass profiles disagree on a shared site. Those
163 profiles have no common extension. Because they are chosen independently
164 of phase, synchrony places no constraint on their disagreement.

165 This distinction limits the biological interpretation. Receptive territories
166 and content domains must be related to spatial patches, and the descriptions
167 must support the required extension. Matching overlap marginals of de-
168 coded probability laws need not determine a unique global joint distribution
169 (Abramsky & Brandenburger, 2011).

170 The cover is itself a choice, and the construction constrains it in one way
171 only: synchronized patches agree where they overlap. Where they do not
172 overlap there is nothing to compare, so a cover by pairwise disjoint patches

173 satisfies that obligation for *every* assignment of local descriptions and phases
174 and the gluing theorem still returns a unique global section, which is the
175 tuple it was handed. Membership of the class then reports nothing about the
176 system beyond the patches covering it. What a gluing claim asks of a cover
177 is therefore a non-degeneracy the theorem does not assume: that distinct
178 patches meet, and that what they share is a quantity fixed independently of
179 the agreement it is being used to establish.

180 The cup example exposes a second gap, between two kinds of overlap.
181 The sheaf is built over a spatial substrate, so its patches overlap where they
182 share sites. Visual and motor territories that agree on cup location overlap
183 in what they describe, not in where they lie, and the construction supplies
184 no relation between the two. One route recasts the construction over content
185 variables: each region becomes a partial decoder whose domain is the set of
186 variables it estimates, and two regions overlap exactly where they estimate
187 a common variable. Gluing and approximate selection hold on such a cover,
188 and there compatibility is a consequence: regions decoding a variable within a
189 tolerance agree on it within twice that tolerance, so only agreement beyond
190 fidelity is evidence. The phase-to-content bounds concern spatial patches
191 and do not transfer. One principle selects such a cover without presupposing
192 the agreement it is used to test. Assign each region the variables that can
193 be decoded from its activity above a declared criterion, on data held out
194 from the agreement test, and let two regions overlap on exactly the variables
195 decodable from both. Decodability is fixed by what each region carries before

any comparison is made, so the cover cannot be chosen to make agreement come out. It is non-degenerate exactly when its nerve is connected, meaning that every region is linked to every other through a chain of shared decodable variables, and recordings can fail that condition. The principle has three inputs, the candidate variables, the criterion and the decoder class, and each has a rule that does not consult agreement.

The candidate variables are those the experiment manipulates and the task makes behaviourally relevant, fixed before recording. Data-driven alternatives, such as directions of maximal shared variance between two regions, select for the agreement under test and are excluded. The criterion becomes a statistical one: a region encodes a variable when its cross-validated decoding exceeds a permutation null at a pre-registered level, and results are reported across a declared range of levels. The level need not be exact, because two accurate decoders agree automatically (Eq. (13)); the statistic of Section 10.2 is agreement in excess of what each region’s accuracy implies, which remains valid at any admitted level. The decoder class is bounded on both sides. Unrestricted nonlinear decoders recover a variable from any region carrying information about it, so nearly every pair of regions overlaps and the cover again constrains nothing. The positive bound of Section 4.2 requires encoders with a declared Lipschitz constant L , which enters δ_r directly, so the class should consist of L -Lipschitz maps; linear readout, the usual operationalization of information explicitly available to downstream neurons (DiCarlo & Cox, 2007; Kriegeskorte & Diedrichsen, 2019), is one such class. Within the

219 admitted family, the decoder and the distance in which content discrepancy is
 220 measured are chosen by held-out decoding accuracy and never by agreement.
 221 Neither is neutral: which similarity measure best describes neural activity
 222 differs across tasks (Bobadilla-Suarez et al., 2020). What remains is that the
 223 cover is the experimenter’s. Whether cortex realizes unity over the chosen
 224 variables is the empirical claim under test, and no choice of decoder settles
 225 it. The correlate is therefore operational relative to these pre-registered in-
 226 puts, and carrying the formal results to such a cover is the recasting sketched
 227 above.

228 **4.2 When coherence does constrain content**

229 A declared relation between phase and content gives a positive result. Let
 230 contents take values in a real normed vector space $\mathcal{C}$. Patch i reads an
 231 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 232 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance,
 233 meaning that an encoder’s output moves by at most L times the chord
 234 between two input phases, and require $e_{i,x} = e_{j,x}$ wherever patches i and
 235 j overlap. Encoders may vary across sites; their equality on overlaps is a
 236 supplied relation between descriptions of the shared quantity. Write $z_k = e^{i\theta_k}$
 237 and $m = N^{-1} \sum_k z_k$, so $|m| = r$. The phasors’ squared deviations from their
 238 mean sum to $\sum_k |z_k - m|^2 = N(1 - r^2)$, and $|z_i - z_j| \leq |z_i - m| + |z_j - m|$
 239 gives $|z_i - z_j|^2 \leq 2N(1 - r^2)$ for each pair. Consequently every overlap

240 discrepancy is bounded by

$$\delta_r = L\sqrt{2N(1-r^2)}. \quad (5)$$

241 Here discrepancy is the norm of the difference, and r and N are read on
242 whichever population the bound is applied to. That choice is what decides
243 whether the estimate says anything. A patch carries an order parameter
244 of its own, and reading the bound there carries that patch's resultant and
245 that patch's site count, so a large population observed through small patches
246 is constrained by the small count. Read instead on the whole array, the
247 factor $\sqrt{N}$ makes the estimate weak, because it draws a pointwise guarantee
248 from an average over N sites. The order cannot be improved in general:
249 one oscillator in antiphase among N leaves $N(1-r^2)$ near 4. At $r=1$ the
250 observations agree exactly at either scale. A shared scalar encoder is a special
251 case. Dropping the overlap relation permits disagreement even at perfect
252 locking. A common nonexpansive linear projection preserves agreement.

253 What Eq. (5) bounds is a chord, which never exceeds 2, so the estimate
254 constrains anything only where the right-hand side falls below that. At the
255 sheet's own patch order, the bound is informative for at most 10 sites. Three
256 conditional results decline the step that puts $\sqrt{N}$ there. A pair count bounds
257 the fraction of disagreeing pairs at any population size but names no pair.
258 A chain across the sites that overlapping patches share compares distant
259 territories through the patches between them, with error growing linearly in

260 hops rather than with population: on the same sheet it stays informative
 261 for 19 hops, 0.312 mm. A spectral gap removes the population count at a
 262 frequency-locked configuration, given a declared algebraic connectivity of the
 263 coupling graph. Territories whose agreement unity concerns lie centimetres
 264 apart, beyond the chain's reach, and across them the coherence-to-content
 265 link is the bridge assumption E78 (Table 1). Supplemental Material gives
 266 the three derivations and the sheet's numbers.

267 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 268 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 269 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 270 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (6)$$

271 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 272 duce an initial mismatch. Approximate observations leave a nondecaying
 273 term in the error bound. Taking a uniform bound on Eq. (5) connects this
 274 update to the phase process.

275 **4.3 What approximate agreement determines**

276 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 277 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 278 Weights that vanish outside their patches and sum to one at each site select
 279 a global profile by weighted averaging. The selection lies within ε of each

280 patch there, and any two such selections differ by at most ε , independently
 281 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
 282 $L\varepsilon$ across selections.

283 At zero tolerance, the selection is the unique exact extension. At every
 284 positive tolerance there are families with no exact extension; selection then
 285 gives an approximation whose dependence on weights is bounded. Supple-
 286 mental Material gives a three-site witness attaining the bound. The mass-
 287 profile construction uses the scalar case of the content dynamics and addi-
 288 tionally requires nonnegativity of the initial profiles and observations.

289 These constructions concern declared objects: a specified cover, contents
 290 valued in a normed space, and encoders related on overlaps. Section 8 sets
 291 out what a measurement of their neural counterparts would have to establish.

292 **5 Self-representation: a state that models it-** 293 **self**

294 **5.1 The reflexive construction**

295 Compatibility gives a global state. Reflexivity relates it to an encoding within
 296 that state. Equip global sections with a complete metric. *Restriction reso-*
 297 *nance* is the additional condition that the encoding is the state’s own restric-
 298 tion to a region $\mathcal{A}$, so that the region encodes the state by being part of it.

299 Under that condition the reconstruction map is

$$F(s) = \text{readout}(s|_{\mathcal{A}}). \quad (7)$$

300 The region and readout are fixed across the state space. If F contracts and
 301 fixes the section s_* formed from the local descriptions, Banach's theorem
 302 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
 303 (Table 1 gives the full composition and its edges) supplies the contraction
 304 and $F(s_*) = s_*$; restriction resonance is separate. Supercritical coupling
 305 makes the stipulated contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ less than one for a
 306 relaxation time $\tau > 0$; Section 11 states the standing of that factor. A fixed
 307 contraction can reconstruct accurately only a restricted family of states: if
 308 its error is at most ε throughout that family, any two members are at most
 309 $2\varepsilon/(1 - \Lambda)$ apart.

310 In the cup example, the global state includes the represented scene and the
 311 perceiver's relation to it. A local encoding must retain enough information
 312 for the readout to recover that modelled state; cup position alone would be
 313 insufficient if hand configuration and viewpoint could not be reconstructed.

314 For the same map with Lipschitz factor $\Lambda < 1$, approximate reconstruc-
 315 tion gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1 - \Lambda}. \quad (8)$$

316 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 317 fixed positive ε and τ . Exact uniqueness survives while the error guarantee

318 weakens.

319 5.2 Reconstruction across a family of states

320 Sensitivity can be required quantitatively. Declare relevant macrostates, an
321 encoder Φ and readout R , and require $d(s, R(\Phi(s))) \leq \varepsilon$ throughout that
322 family. If two states have the same code, the triangle inequality gives

$$\Phi(s) = \Phi(t) \implies \max\{d(s, R(\Phi(s))), d(t, R(\Phi(t)))\} \geq \frac{1}{2}d(s, t). \quad (9)$$

323 Thus M relevant states pairwise more than 2ε apart require at least M
324 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
325 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
326 existing self-prediction map.

327 This is an additional accuracy criterion on a family, with no requirement
328 for multiple exact fixed points. For the same fixed contracting map used by
329 E89, however, the diameter bound above is an additional restriction: at zero
330 error the family contains at most one metric-distinct state. A broad family
331 needs an input-conditioned map, one contraction F_u per input u . If F is
332 L -Lipschitz in u , members with inputs u, v lie within $(2\varepsilon + L d(u, v))/(1 - \Lambda)$
333 and the fixed points within $L d(u, v)/(1 - \Lambda)$, so a self-state can follow the
334 scene. The code criterion then binds within each input: a readout storing
335 one state per input reconstructs without encoding anything. Under the fixed
336 map of E89 the minimal-self correlate is a condition on one self-state. A blind

337 encoder or constant readout fails the criterion for separated states despite
 338 having a unique fixed point.

339 Where the encoding is read as the state’s own restriction to a region, the
 340 criterion becomes a condition on the region. Restriction resonance requires
 341 the encoder and that restriction to identify the same pairs of states, so a
 342 boundary reconstructing two relevant states more than 2ε apart while its
 343 region reads them identically is not resonant. Read forwards, a resonant
 344 boundary that reconstructs a pairwise separated family restricts injectively
 345 on it. Which states a region reads identically is fixed by where the region
 346 sits, before an encoding is chosen, so the resolution the family demands is
 347 the region’s and no readout supplies it. The implication runs one way: a
 348 region that does resolve the family is not thereby shown to carry a resonant
 349 encoding.

350 Resonance and resolution are jointly satisfiable without a local fold. Take
 351 the region to be the whole substrate and use the identity readout: restric-
 352 tion is the identity, so this boundary is resonant and reconstructs every de-
 353 clared family exactly. Its map has Lipschitz factor one and does not satisfy
 354 E89’s strict contraction premise on a nontrivial metric space. A properness
 355 condition on the region is therefore a separate requirement, and the fixed-
 356 contraction diameter bound remains in force when E89 is also required. With
 357 an L -Lipschitz readout, a family reconstructed within ε whose members lie
 358 pairwise more than $d_{\min}$ apart has δ -separated codes whenever $\delta L \leq d_{\min} - 2\varepsilon$,
 359 so the family is no larger than the δ -packing number of what the region car-

360 ries. At the whole substrate it says nothing, because the region's range is
 361 everything. The gain L is a declared input, and the bound weakens without
 362 limit as the gain grows.

363 **5.3 Agreement after a change takes communication**

364 Snapshot compatibility does not specify how quickly descriptions respond
 365 when a shared quantity changes. In a synchronous network whose updates
 366 read only the site's own state and declared neighbours' messages, two initial
 367 configurations agreeing on a site's causal past through T rounds produce the
 368 same state there at round T . The deadline consequence is the one Section 2
 369 states, and it follows from the reconstruction bound in Section 5.2: a report
 370 from that site is a readout of a code that the two interventions leave identical.

371 A three-site delay line witnesses failure after one round and exact report-
 372 ing after two; cutting the path prevents delivery at every round. A report
 373 may still happen to be correct about one fixed world. The bound concerns
 374 responsiveness across interventions.

375 Throughout this section *prediction* names a reconstruction map, and the
 376 relevant family of states, the metric, the tolerance, the communication graph
 377 and the deadline are all declared inputs, as are the represented variables and
 378 the fidelity counted as accurate. Reading the map as temporal neural pre-
 379 diction would require dynamics; reading the resulting code count as physical
 380 memory capacity would require a hardware model, which a state-dependent
 381 readout would have to extend.

382 6 What the conditions ask of a digital candi- 383 date

384 A claim that a digital system, such as a large language model (LLM) run-
385 ning on graphics processing units (GPUs), has unified content or represents
386 itself commits its maker to the measurements this account asks of cortex.
387 The maker must identify the shared quantities the system’s parts describe,
388 show that decoders fitted independently on those parts agree about them,
389 and show that a declared local code separates the relevant family of states
390 and survives intervention. The indicator-properties approach assesses arti-
391 ficial systems against computational properties drawn from theories of con-
392 sciousness (Butlin et al., 2023); the conditions here are fewer and each is
393 a measurement rather than an architectural property. A digital system is
394 also where they can be measured today, with full access to its state and
395 exact interventions on it: fitting decoders for one shared quantity on two
396 separate layers, heads or token positions, and testing their agreement and
397 whether reconstruction survives intervention, would carry out the proposed
398 measurement directly.

399 One constraint follows from the architecture alone. Within one forward
400 pass of a causally masked model, position t reads only its causal past, so an
401 intervention outside that past cannot be reconstructed at t by that step. This
402 is the deadline result of Section 5 applied to the mask, and it bounds which
403 variables a position can encode about the rest of the sequence within one

404 step. A second constraint needs a premise the architecture does not supply:
 405 a linear encoder and readout factoring through d coordinates pay reconstruction
 406 error at least the discarded spectral tail of the coupling operator, but
 407 query/key dimension does not supply such a bottleneck for softmax attention.
 408 These results rank no architecture family, imply no general inferiority of
 409 GPU hardware and settle no question of machine experience. Supplemental
 410 Material gives the bottleneck derivation, a synthetic two-view task with its
 411 controls, the information ceiling of token outputs and the hardware scope.

412 **7 The formal composition**

413 The Lean development composes the three requirements into one conditional
 414 theorem. Its passive and feedback branches each accept eight connecting hy-
 415 potheses, running from finite capacity through a register, dissipation and a
 416 continuum coupling to the field, coherence, the cover and the self-prediction
 417 fixed point. Each edge is a hypothesis of the composition, and its inferen-
 418 tial content is its last step, where contraction and a fixed glued section give
 419 uniqueness. Table 1 lists the four edges the article argues from; Supplemen-
 420 tal Material gives the full chain, its thermodynamic edges and the register
 421 ledger of the feedback branch. Both branches also take a cover carrying ex-
 422 act overlap compatibility and a reflexive boundary specifying encoding and
 423 readout, and conclude that the glued section is the unique fixed point of self-
 424 prediction. Composing the edges establishes that the premises are consistent,

425 named in one place and machine-checked together, and joint witnesses satisfy
426 every edge of each branch at once. The development declares no additional
427 axioms beyond its libraries' foundational axioms.

Table 1: The four connecting hypotheses the article argues from, shared by both branches of the formal composition. Supplemental Material gives the full eight-edge chain and, in Table S1, maps the node-level results.

Connection	Status	Required assumption
E56 <i>field identification</i>	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67 <i>above threshold</i>	Physical commitment	That field operates above the synchronization threshold.
E78 <i>coherence to cover</i>	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements. Patches and oscillators carry one index, so one oscillator carries exactly one content patch: a modelling assumption rather than a consequence, and not an innocent one, since field-mediated coupling is spatial while two territories describing one quantity need not be neighbours. Where the cover's nerve is connected the same edge carries through the sites its patches share, at a residual linear in that nerve's diameter; where the configuration is frequency-locked it carries through the coupling graph's algebraic connectivity λ_2 instead of through any patch count, which asks the cover for nothing and is declared hardware data in the standing of κ .

Table 1 (continued)

Connection	Status	Required assumption
E89 <i>contraction to fixed point</i>	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $\Lambda < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.

8 What the observations establish

The composition holds under its premises. Whether observation can establish those premises is a separate question, and the three requirements it connects concern different objects: phase dynamics, local descriptions and internal encodings. Their observables must retain these distinctions. The following controls show which mechanisms remain indistinguishable, which temporal approximations fail, which states one observable cannot separate, and what independent content measurements must establish. Each is a finite numerical regime on a declared model, with full parameters and controls in Supplemental Material, and none of them calibrates a cortical quantity.

8.1 Stationary distributions leave the mechanism undetermined

Equation (2) gives a concentration-coherence relation with no fitted shape parameter. Estimate r from circular resultant length and a from density shape: inverting R from the measured r would make agreement automatic.

443 Even independently estimated agreement identifies a statistical family shared
 444 by different mechanisms.

445 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
 446 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
 447 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
 448 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
 449 and diffusion D_0 have that same density. Their stationary distributions co-
 450 incide window by window despite different causes.

451 Resolved dynamics supply additional information. In the ideal model
 452 with continuous latent paths, quadratic variation accumulates at rate $2D$
 453 and conditional increment means constrain drift; a filtered observation of
 454 the phase needs its own observation model before either is read off. Separat-
 455 ing endogenous coupling from a matched common drive additionally requires
 456 a measured drive or controlled perturbation; identifying electromagnetic cou-
 457 pling requires evidence specific to that interaction. Supplemental Material
 458 gives the equivalence construction and its numerical check. Phase statistics
 459 pooled over time and over bipolar pairs cannot supply these estimates: on
 460 public sedation EEG (Bajwa et al., 2025), the pooled pipeline places a per-
 461 fectly phase-locked signal at $r = 0.0001$, so a spatial test needs an estimand
 462 that removes the common rotation and a montage whose effect on spatial
 463 phase is modelled; Supplemental Material reports this estimator exercise and
 464 the dependence-aware design.

465 **8.2 Finite ramps change the observable onset**

466 The square-root law in Eq. (3) describes stationary solutions, and a changing
467 coupling needs time for the phase distribution to approach them. In a deter-
468 ministic mean-field control crossing the threshold, none of the scanned rates
469 meets the declared tracking tolerance at onset. In a finite-population ramp
470 study ($N = 2000$, $D = 1$, 32 replicas, 8 speeds), faster ramps delay escape.
471 The fitted delay exponent 0.444 has a replica-bootstrap interval $[0.398, 0.496]$
472 that excludes the ideal $1/2$, a deterministic reference shows that the escape
473 criterion changes the fitted exponent, and no common asymptotic exponent
474 is established. Delayed passage through a noisy bifurcation (Berglund &
475 Gentz, 2002) makes the apparent exponent depend on ramp speed and fit-
476 ting window. Supplemental Material gives the sweep and its figure, the
477 fluctuating-coupling and population-size controls, and the signed-network
478 crossing bracket.

479 **8.3 A winding state leaves the order parameter am-** 480 **biguous**

481 The global order parameter r discards spatial arrangement. On a periodic
482 sheet, a phase field can wind once around the sheet and have $r = 0$ while
483 neighbouring sites remain locked. At identical frequencies, every such wind-
484 ing is stationary for an isotropic symmetric kernel: contributions at opposite
485 separations cancel. It violates the quarter-turn initial-spread premise of the

486 convergence result, which is why that result can still select the uniform state.

487 In the spatial control, the winding persists below the interpolated de-
 488 cay length 0.1871 mm and unwinds above it (Fig. 3). At 0.1 mm, uniform
 489 and winding starts under the same coupling, diffusion and kernel end at
 490 $r = 0.9657$ and $r = 0.0083$; patch-local order is 0.9015. Under quenched fre-
 491 quency spread 1.5 rad s^{-1} , the measured retention boundary is 0.1421 mm,
 492 and the corresponding order comparison is 0.9418 against 0.0348 over 3 frozen
 493 realizations. These are finite controls, not a cortical wave identification or a
 494 derived transition.

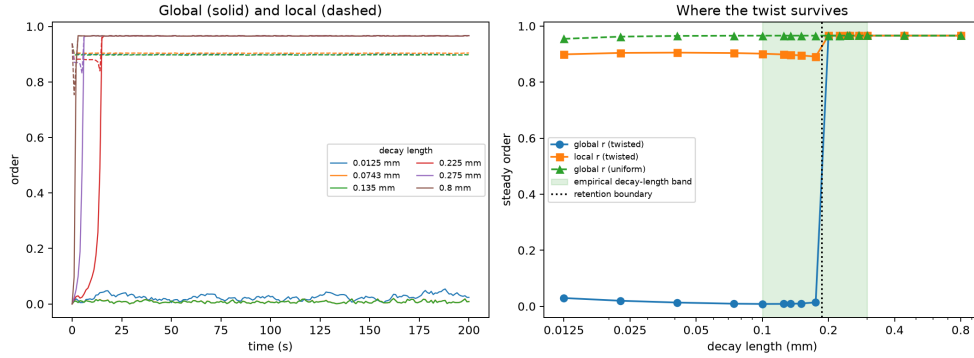


Figure 3: A winding phase field on the spatial sheet, at identical frequencies. Left: global order (solid) against patch-local order (dashed) across a subset of coupling decay lengths. Right: the twisted and uniform initial conditions compared across lengths, with the shaded 0.1–0.3 mm band and the interpolated retention boundary. Below that boundary the two initial conditions reach different stationary values of r at the same coupling, diffusion and kernel, so the resultant does not report the coupling. The sweep is sampled at the lengths marked.

495 Linearization about a winding gives explicit mode rates, which concern
 496 linear stability rather than the measured retention boundary. An amplitude

extension adds an existence condition under which a broad kernel can remove a winding that the phase-only equations retain as a stationary state.

Neither model turns these states into measured cortical spirals. The periodic sheet cannot carry a lone phase defect, the relaxational dynamics has no propagation speed, and the amplitude variable has no calibrated relation to a recorded envelope. Supplemental Material gives the spectral, amplitude and defect results. Locally, Eq. (5) can still bound content disagreement on a winding patch when shared encoders are supplied, although its global $r = 0$ form is uninformative. A nonzero ring winding number q also obstructs a global real-valued phase lift; it has no identified content meaning here. Thus an onset criterion based only on r can register a change of basin as well as a change of coupling. The protocol of Section 9.3 needs a spatial phase observable.

8.4 Coherent adaptation can lose environmental alignment

An adaptive coupling rule might supply representational structure, but its objective must reward the relevant agreement. For symmetric coupling, pairwise sine terms cancel in the population mean drift. Minimizing squared drift at that fixed mean therefore rewards equalizing drift across nodes. In the frequency-cluster control, this favors coupling between unlike-frequency clusters, whereas the diagnostic environmental template rewards coupling

518 within clusters. Supplemental Material proves descent of this objective for
519 a specified gradient flow at fixed phases and distinguishes it from entropy
520 production.

521 The finite control co-evolves noisy phases and symmetric, nonnegative
522 couplings at fixed total resource. Its template is a diagnostic covariance
523 of latent signals, never a training input. Across three illustrative seeds,
524 second-half order remains between 0.9167 and 0.9222, while the within-
525 cluster to between-cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–
526 0.6135 (Fig. 4). Random-update, permuted-gradient and frozen-kernel con-
527 trols are reported in Supplemental Material.

528 The immediate post-update objective reduction disappears when aver-
529 aged over complete update intervals. A prespecified grid of 6 rate–cadence
530 combinations contains no case meeting the 5% sustained-reduction criterion
531 while retaining coherence and losing alignment. The finding is coherent adap-
532 tation away from the specified structure, with no demonstrated sustained cost
533 reduction.

534 Kernel alignment and overlap compatibility are different observables.
535 This control addresses acquisition of its designated template under its par-
536 ticular objective. Learning models can explicitly include data in an objective
537 (Corberi et al., 2026); an account of agreement about the cup must specify
538 the content its learning rule is meant to preserve.

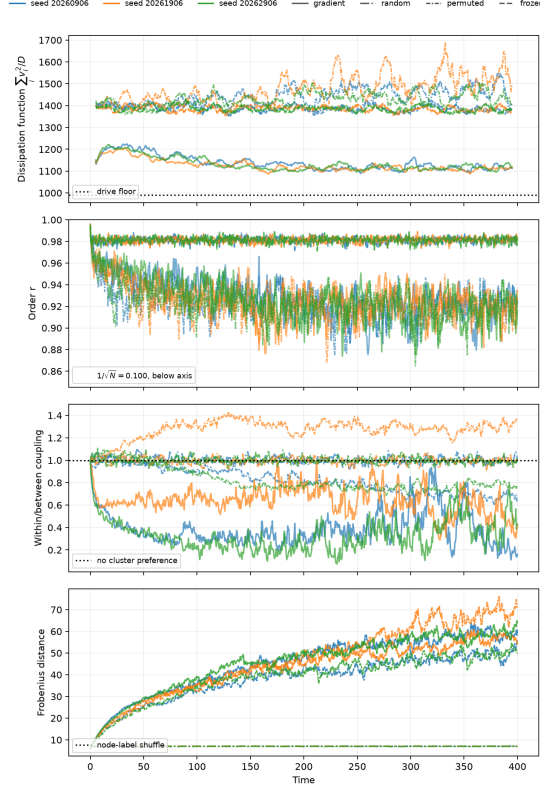


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), permuted gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

539 8.5 Content agreement needs independent decoders

540 A neural test needs independently identified domains, local descriptions and
541 restriction maps. Two decoders should estimate a shared content on over-
542 lapping domains before their agreement is examined. Discrepancy requires a
543 null under decoding noise and controls using deliberately compatible and in-
544 compatible examples. An implementation on synthetic data separates these
545 cases at $\text{AUC} = 0.986$, on a 95% interval of $[0.980, 0.992]$ resampled by trial;
546 the compatible construction scores 0.476 on an interval containing one half.
547 Shared-prior shrinkage can nevertheless manufacture agreement, and regional
548 bias can manufacture disagreement. Per-patch decoding diagnostics detect
549 these failures; the compatibility statistic alone does not.

550 On constructed content, phase carries a share 1.000 of the content entropy
551 when content is built from it and 0.000 when it is not. A phase-derived statis-
552 tic reports compatibility on phase-locked patches whose local descriptions
553 disagree. The implemented content statistic detects their incompatibility.
554 These controls validate a measurement on known constructions; a neural
555 content model and decoder remain to be established.

556 Choosing the cover also chooses a candidate subject: trivial descriptions
557 can agree while carrying little content, and a partition can conceal disagree-
558 ment. Section 4 gives the decodability rule that fixes the cover without
559 consulting agreement.

560 **9 The cortical hypothesis and an awakening** 561 **test**

562 The cortical proposal identifies a physical source of coupling and asks whether
563 its changes help explain recovery. The tests in Section 8 determine what
564 observations must distinguish before a phase trace can support this identifi-
565 cation.

566 **9.1 Physical identification and calibration**

567 The cortical hypothesis identifies part of effective coupling with endogenous
568 extracellular electromagnetic interactions. Fields can modulate spike timing
569 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
570 propose coordination of distributed neural information (Pinotsis & Miller,
571 2022; McFadden, 2020). Phase organization under propofol motivates study-
572 ing this regime (Bastos et al., 2021).

573 The formal field predicate requires a jointly continuous kernel, a normal-
574 ized substrate, coupling equal to the kernel average, and no single site pro-
575 viding more than half that strength. Toy constructions satisfy it. A cortical
576 realization must specify the scale at which the kernel and its reduction ap-
577 ply: calcium imaging reports different state-dependent modularity patterns
578 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

579 Cortical columns are a serious candidate for the oscillator index. They are
580 delineated by response properties and anatomy rather than by the agreement

581 they would be used to establish (Mountcastle, 1997), and a phase can be
582 estimated from a column-sized territory at the electrode spacings used for
583 cortical wave measurements (Townsend et al., 2015). They are a poor content
584 cover: non-overlapping columns form a disjoint cover (Section 4), much of
585 association cortex has no accepted columnar parcellation, and the column’s
586 standing as a canonical unit is contested (Horton & Adams, 2005; Rakic,
587 2008). The content cover is the decodability cover of Section 4, and the
588 bridge assumption E78 records its identification with the oscillator index.

589 The spatial reduction is also testable. A row-sum aggregation at zero
590 diagonal tracks the uniform-coupling control over the tested spatial decay
591 range, while frequency heterogeneity displaces the operational threshold. The
592 fit window biases both estimates relative to $2D$; agreement with the control
593 is therefore the supported conclusion. Supplemental Material reports the
594 reduction, its numerical error and the distinct assumptions needed to identify
595 the kernel average with the scalar coupling of the installed-energy condition.

596 Converting a spike-time shift into coupling needs a rate: the dimensionless
597 product of population size N , timing sensitivity $\mathcal{T}$, field amplitude E and
598 frequency f becomes a coupling through a conversion rate γ ,

$$K_{\text{eff}} = \gamma N \mathcal{T} E f. \quad (10)$$

599 This rate describes how induced timing shifts accumulate into drift.

9.2 Extracellular geometry and recovery

The proposed control variable is extracellular geometry. Extracellular volume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková & Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density, an effective-medium description predicts that a larger extracellular fraction raises conductivity and reduces field amplitude.

The coupling-gated hypothesis consequently predicts that extracellular contraction on awakening raises E and, holding the other factors in Eq. (10) fixed, raises effective coupling. This argument supplies a direction; the geometry–coupling relation, its magnitude and its role in human recovery require measurement.

For a smooth crossing at nonzero speed, evaluating the stationary branch along the changing control gives a deterministic tracking approximation that fails at onset in the tested crossing control:

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (11)$$

This is a deterministic tracking approximation from Eq. (3) at fixed D ; the crossing controls in Section 8.2 show that it fails as an observed-onset approximation under the tested initialization. A separation of geometry and phase time scales can support tracking away from threshold. The awakening

hypothesis needs a calibrated dynamical prediction for an operational onset
above the finite-size fluctuation floor. Crossing K_c alone does not supply a
recovery latency.

9.3 A protocol for the awakening experiment

Neural inertia describes a behavioural barrier and hysteresis between anaes-
thetic induction and emergence (Friedman et al., 2010); recovery can pass
through metastable activity states (Hudson et al., 2014; Proekt & Hudson,
2018). The $v^{1/2}$ exponent of an ideal delayed bifurcation is generic (Baer
et al., 1989; Berglund & Gentz, 2002) and cannot alone discriminate a field
mechanism from a synaptic one. A possible discriminator is spatial onset
relative to independently estimated local field coupling and network con-
nectivity. The field-based onset map below is a separate model from the
mean-field one, specified before the observations are inspected.

For that comparison, partition the recording into prespecified regions i
and measure pre-onset order $r_i(t_0)$ and diffusion D_i . Define a field score
 $h_i = N_i \mathcal{T}_i E_i f_i$ from independent pre-onset field and timing measurements,
and a connectivity score c_i from a separately estimated structural network.
For the same measured emergence drive $g(t)$, compare $K_i^F(t) = a_F g(t) h_i$ with
 $K_i^C(t) = a_C g(t) c_i$. Each model has one fitted scale. The local first-harmonic
approximation predicts

$$\hat{r}_i^M(t) = r_i(t_0) \exp\left(\int_{t_0}^t [K_i^M(s)/2 - D_i] ds\right), \quad M \in \{F, C\}. \quad (12)$$

640 Predicted onset is the first crossing of a region-specific threshold fixed from
641 pre-onset fluctuations. Fit the two scales on training sessions, then com-
642 pare held-out onset times and site order under the same observation model.
643 Larger field scores predict earlier onset only when baseline order and diffu-
644 sion do not reverse the ranking. This declared approximation omits interre-
645 gional coupling and saturation; a successful ranking would support its field
646 covariate, not prove ephaptic causation. Supplemental Material specifies the
647 controls and failure conditions.

648 A within-subject protocol can manipulate emergence rate, with spatially
649 resolved recordings to compare the proposed field and connectivity models:

- 650 1. **Manipulate emergence rate.** Vary the propofol infusion taper
651 across sessions within the same subject, recording drug concentrations
652 or a validated pharmacokinetic proxy to quantify the manipulation.
- 653 2. **Measure behavioural recovery.** Record time-to-response to a stan-
654 dard stimulus as the independent behavioural endpoint, rather than
655 estimating ΔK directly.
- 656 3. **Resolve spatial onset.** Identify anatomical locations where coher-
657 ence first crosses the baseline-scaled threshold and score the two maps
658 of Eq. (12) on held-out sessions. The two covariate maps are fixed
659 before the outcome sessions are inspected.
- 660 4. **Match physical controls.** Confirm that the observed coherence-to-
661 amplitude relation holds under the specific local diffusion D and ob-

662 servation noise. The model’s quadratic-variation rate $2D$ applies to
663 the latent phase; a band-limited analytic phase is smooth below its
664 correlation time, and on a constructed field its squared-increment rate
665 changes 37-fold between 0.2-ms and 20-ms steps. Estimating D there-
666 fore requires an explicit observation model for filtering and sampling.

667 Predeclare the recording window, regional fluctuation thresholds, and a
668 shared timing-error model. Fit the two scales and that observation model
669 on training sessions only; on held-out sessions, score the observed site-onset
670 times and sites that never cross by the same regional predictive rule. Report
671 site order as a secondary outcome and behavioural time-to-response sepa-
672 rately. Sessions without a crossing are censored observations, not zero delays
673 or exclusions. Compare the two scores only where independently estimated
674 field and connectivity maps predict distinguishable onset patterns at the
675 recording resolution. A held-out field-model advantage would support that
676 specified spatial predictor; it would not identify an ephaptic cause, shared
677 content or experience. Supplemental Material gives the controls and pilot
678 quantities needed to judge whether the comparison is informative.

679 10 Discussion

680 10.1 Unity and a minimal self as interpretive motiva- 681 tion

682 We propose the state glued from compatible local descriptions as a formal
683 correlate of conscious unity, and its accurate internal reconstruction as a cor-
684 relate of a minimal self. In the cup example, this would mean one description
685 that preserves the cup’s relation to the perceiver as well as the scene. Com-
686 patibility permits distinct contents and uncertainty; the shared quantities
687 must be chosen from independent evidence about what each region encodes.
688 The reconstruction need not be a verbal or autobiographical account.

689 The minimal self, in Gallagher’s sense, is the pre-reflective sense of be-
690 ing the subject of an experience, present before any narrative or reflective
691 self-description (Gallagher, 2000). Two of its features motivate the proposed
692 correlate. It is not a separate object of experience but a perspective built
693 into the experienced scene; correspondingly, the reconstruction is not a sec-
694 ond representation beside the glued state but a map whose fixed point is
695 that state, read through a part of it. It is immediate rather than inferred
696 from a higher-order state; correspondingly, the fixed point needs no further
697 state that represents the first. The correspondence is structural. Self-model
698 theory locates the phenomenal self in a transparent model a system has of
699 itself (Metzinger, 2003), and interoceptive-inference accounts ground it in
700 prediction of the body’s own states (Seth, 2013). The present condition is

701 narrower than either: it asks only that a declared local encoding recover the
 702 global state, including perceiver-relative variables, at a declared fidelity, and
 703 it says nothing about transparency or about which bodily variables must be
 704 modelled. Either account could supply the family of states and the fidelity
 705 that the condition leaves as inputs. Under E89 that family is one state at
 706 zero error and a family of diameter at most $2\varepsilon/(1 - \Lambda)$ within error ε (Sec-
 707 tion 5.2), so under E89 the condition constrains a single self-state; a self that
 708 persists through change needs the input-conditioned map of Section 5.2. Its
 709 proposed relation to experienced perspective remains an interpretation. Nei-
 710 ther identification is among the connecting hypotheses of the composition
 711 (Table 1), and satisfying them would not measure experience. The unity
 712 proposal is also only as definite as its cover. The gluing theorem holds for
 713 whatever cover it is given, a disjoint cover satisfies it trivially, and the cover
 714 of interest overlaps in content rather than in space. Section 4 proposes fixing
 715 that cover by decodability on held-out data, which excludes covers chosen for
 716 agreement and makes non-degeneracy a testable property of the recordings.
 717 Its variables, criterion and decoder class are pre-registered by rules that never
 718 consult agreement.

719 10.2 A comparison that could disconfirm the link to 720 content

721 The framework predicts that conscious content tracks compatibility, not co-
722 herence. The comparison must also separate compatibility from how well
723 each region encodes the stimulus, because two accurate decoders agree auto-
724 matically. If decoders d_A and d_B estimate a shared stimulus quantity x with
725 errors $e_A = d_A - x$ and $e_B = d_B - x$, then

$$|d_A - d_B| \leq |e_A| + |e_B|, \quad |d_A - d_B|^2 = |e_A|^2 + |e_B|^2 - 2\langle e_A, e_B \rangle. \quad (13)$$

726 Consciously perceived stimuli are usually decoded better, so higher agreement
727 on those trials would follow from single-region fidelity alone. At matched
728 single-region error, agreement differs only through the covariance of the two
729 errors, which is the term compatibility bears on: two regions describing one
730 content should err together.

731 Present matched stimuli that are perceived on some trials and not on
732 others, for example at masking or rivalry threshold, and fit decoders for one
733 shared stimulus quantity independently on two regions that both encode it,
734 selected by the decodability criterion of Section 4 on separate data. Clas-
735 sify trials by perceptual state with a no-report measure, such as reflexive
736 eye movements or pupil responses that track the dominant percept in rivalry
737 (Tsuchiya et al., 2015), so that the classification does not also select for the
738 neural correlates of reporting; report-based classification is a secondary anal-

739 ysis. Match the two classes on phase coherence between the regions and
740 on each region’s own decoding error (or, where the stimulus value is uncer-
741 tain, on each region’s signal strength). The statistic is the excess agreement
742 over a fidelity-matched null: pair region *A*’s decode on one trial with region
743 *B*’s decode on another trial of the same stimulus and error stratum, which
744 preserves both marginal accuracies and removes any coupling of the errors.

745 The proposed link predicts a larger excess on perceived trials at matched
746 coherence and matched single-region accuracy (Fig. 5). Equal excess across
747 classes, or excess that follows coherence and not perceptual state, counts
748 against it. A positive result is not yet specific: noise shared upstream of both
749 regions also correlates their errors, so the excess should be compared against
750 an upstream region’s decoding error, matched in the same way. A self-model
751 comparison has the same form: at matched accuracy for the scene variable,
752 the reconstruction of perceiver-relative variables, such as hand position rel-
753 ative to the target, from a declared local encoding should be more accurate
754 on perceived trials. These are correlational tests of proposed correlates, and
755 the no-report classification is itself an indirect marker of experience.

756 **10.3 What a test of the full proposal must establish**

757 A test needs separate estimates of effective coupling and diffusion, shared
758 content on overlapping regions, and reconstruction from a specified encoding
759 at a declared fidelity. The cover, relevant state family and decoder must
760 be fixed without selecting them for agreement. An independent experiential

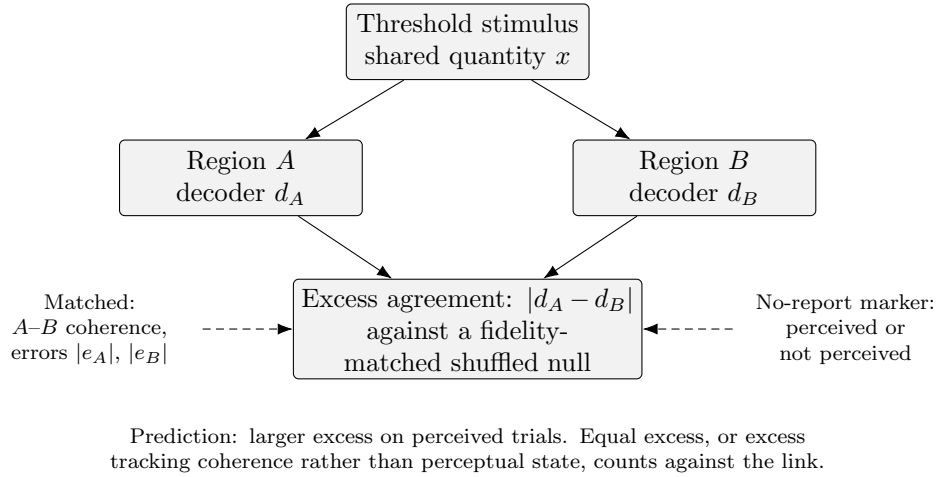


Figure 5: Design of the disconfirming comparison. Decoders for one shared stimulus quantity are fitted independently on two regions. Trials are classed by a no-report marker of perceptual state and matched on inter-regional phase coherence and on each region’s own decoding error, so that agreement cannot follow from single-region fidelity (Eq. (13)). The statistic is agreement in excess of a null that pairs decodes across trials within the same stimulus and error stratum.

761 endpoint is needed to evaluate the interpretation; absence of a report alone
 762 does not define absence of experience. A unified episode with incompati-
 763 ble decoded descriptions would challenge the unity identification. A unified
 764 episode lacking the proposed self-encoding would challenge the minimal-self
 765 claim. Conversely, satisfying the mathematical conditions without such an
 766 episode would challenge their sufficiency as correlates.

767 For the physical route, cortical actuator modes and prices, the geometry–
 768 coupling relation and a dynamical onset model remain to be measured. The
 769 field/connectivity comparison in Section 9.3 evaluates one spatial onset pre-
 770 diction. Controlled changes in coordination and independently decoded con-

771 tent agreement would address the representational connection. The synthetic
772 decoder exercise calibrates part of these measurements, and the EEG exercise
773 identifies the estimand a spatial test needs.

774 Integrated-information, global-workspace and predictive-processing ac-
775 counts organize unity, access and modelling differently (Tononi et al., 2016;
776 Baars, 1997; Dehaene et al., 2014; Friston, 2010). This account specifies con-
777 ditional relations among coupling, local content and reconstruction, each tied
778 to a measurement.

779 **11 Limitations**

780 *Experience and interpretation.* Synchrony is not itself an experiential mea-
781 sure, and nothing here establishes a consciousness criterion: the conditions
782 are proposed correlates to be tested against report or another independent
783 endpoint. The mathematical states are spatial measures and fields, and iden-
784 tifying them with perceptual content requires independent decoding evidence.
785 The constructions of Section 4 select neither the neural cover nor the decoder,
786 and identifying their contents with perceptual content, or establishing that
787 the identification bears on experience, is a further modelling task. The cor-
788 respondence between the reconstruction’s fixed point and the minimal self is
789 structural and does not show that a fixed point is experienced as a perspec-
790 tive. The remaining explanatory question is whether these relations occur in
791 neural activity and matter for experience; the framework does not explain

792 why satisfying them would be accompanied by experience.

793 *Formal scope.* The formal results are conditional on declared encoders,
794 covers and maps whose cortical counterparts are unmeasured. Each edge
795 of the composition is supplied rather than derived from the one above it,
796 so the composition does not establish that self-representation follows from
797 coherence. The joint witnesses satisfy every edge in a specified model: they
798 prove neither that cortex satisfies those hypotheses nor that the glued state
799 is experienced, and establish no common cortical mechanism. The contrac-
800 tion factor Λ is stipulated, not derived. It borrows the rate $(K - 2D)/2$ at
801 which the first harmonic of the uniform density grows (Section 3), a rate
802 about a different state from the one F acts on, and no result derives a Lips-
803 chitz law for an encoding or readout from the phase dynamics; constructing
804 a map with that factor remains a modelling obligation. No physical input-
805 conditioned map F_u is constructed, the gain L of Section 5.2 is derived here
806 for no physical readout, and E89 supplies no resolution of a declared family
807 by the encoding region. Agreement of a common projection does not estab-
808 lish agreement of the full vector, and the general vector result of Section 4
809 supplies no identification with the spatial-measure sheaf. The cover on which
810 compatibility is assessed is fixed only relative to pre-registered variables, de-
811 coder class and criterion, and the phase-to-content bounds are stated for
812 spatial covers rather than for the content cover that criterion yields.

813 *Physical identification.* The formal field condition requires a continuous
814 kernel, a normalized substrate, a specified average coupling and noise, and a

815 site-contribution bound. Those conditions do not identify an electromagnetic
 816 substrate; the extracellular interpretation needs the separate calibration of
 817 Section 9, and the field commitment requires physical evidence beyond a
 818 choice of mathematical representation. Phase organization under propofol
 819 does not measure coupling, and the support comparisons in Supplemental
 820 Material concern specified support constraints and supply neither that evi-
 821 dence nor energy prices. Neither the conversion rate γ nor the installation
 822 factor κ follows from a heat bound, and no cortical value of κ is known.
 823 Testing the physical threshold requires independent estimates of the inter-
 824 action kernel, its aggregation and diffusion in the same cortical regime. No
 825 measured connectivity yet supplies the spectral route of Section 4.2. The
 826 mean-field model has no spatially varying κ and derives no nucleation or-
 827 der, which is why the onset comparison of Section 9.3 rests on a separately
 828 specified model.

829 *Numerical and empirical controls.* The controls of Section 8 are finite
 830 regimes on declared models and calibrate no cortical quantity. A slow con-
 831 trol variable supplies no general guarantee of branch tracking; the winding
 832 sweep establishes no transition and no biological identification; phase agree-
 833 ment cannot replace an independently decoded content measurement. The
 834 synthetic decoder exercise validates a measurement on known constructions,
 835 and the EEG exercise checks a measurement pipeline: it identifies what a
 836 pooled bipolar estimand cannot measure, and neither exercise identifies a
 837 neural content code, tests experience or validates any experiential claim.

838 12 Conclusions

839 Phase coherence, compatibility and self-reconstruction are separable condi-
840 tions. Coherence constrains shared content only through encoders shared on
841 overlaps and with a bound that grows as the square root of the population
842 read. That bound is vacuous beyond 10 sites at the simulated sheet’s patch
843 order; chaining it through overlapping patches reaches 0.312 mm of that
844 sheet, and a spectral route removes the population count only for a declared
845 connectivity at a frequency-locked configuration. Across the centimetre sep-
846 arations of distinct cortical territories, the positive link from coherence to
847 content is the bridge assumption E78 rather than a derived result. Com-
848 patible local descriptions, not synchrony, determine one global state; and a
849 contracting self-map fixes that state while reconstructing only a family of
850 bounded diameter. Each result replaces an assumption about synchrony by
851 a measurement: independently decoded agreement on overlaps, a declared
852 encoding and its fidelity, and calibrated coupling and diffusion. The compar-
853 ison of Section 10.2 states how such measurements could count against the
854 proposed link to conscious content.

855 Notation

Table 2: Symbols used throughout, with the section that defines each. Where one letter carries more than one meaning, the meanings are separated by the section they are read in.

Symbol	Defined	Meaning
θ_i, ω_i	Eq. (1)	Phase and natural frequency of oscillator i
N	Eq. (1)	Number of oscillators in the population the quantity is read on
D	Eq. (1), Supp.	Phase diffusion in inverse time; in the Supplemental Material’s convergence proof, the phase spread $D \leq \pi/2$
K_{ij}, K	Eq. (1)	Pairwise and scalar mean-field coupling, $K_{ij} = K/N$
K_c	Eq. (2)	Synchronization threshold, $K_c = 2D$
r, r_N	Eq. (1)	Resultant length (order parameter), on the population it is read on
a, ψ	Eq. (2)	Concentration $a = Kr/D$ and collective phase of the stationary density
q	Section 8.3	Winding number of a phase field read around a ring
U_{inst}, κ	Section 3.2	Installed energy of a declared actuator and the mode-data factor bounding its mean coupling by κU_{inst}
L	Section 4.2	Lipschitz constant of an encoder or readout
ε, δ	Section 4.2	Content and observation discrepancy tolerances
M, b	Eq. (9)	Size of a declared family of states and code length in bits
Λ, τ	Eq. (7)	Contraction factor $\Lambda = e^{-(K-2D)\tau/2}$ and its relaxation time; in Eq. (4), τ is the duration of a change
J, σ_J, C	Eq. (4)	Probability current of the phase density, its current functional and a current budget
T	Section 5	Number of communication rounds to a deadline
ν	Supplemental Material	Fan-in of a communication graph: the most sites any one site hears from
$\gamma, \mathcal{T}, f$	Eq. (10)	Field-to-coupling conversion rate, timing sensitivity and frequency
α	Section 9	Extracellular volume fraction

Table 2 (continued)

Symbol	Defined	Meaning
λ	Section 9	Extracellular tortuosity
Ω, λ_2	Section 4.2	Common rotation rate of a frequency-locked configuration, and the algebraic connectivity of the coupling graph
β, k	Section 4.2	Chord diameter of a patch and the diameter of a cover’s nerve, in hops
$\mathcal{C}, e_{i,x}$	Section 4.2	Content space and the phase encoder patch i reads at site x
Φ, R	Eq. (9)	Encoder and readout of the reconstruction criterion
$\mathcal{A}, F$	Eq. (7)	Encoding region and the reconstruction map $F(s) = \text{readout}(s \mathcal{A})$
E	Eq. (10)	Field amplitude, in the cortical identification
s, s_*	§4–5	A global section and the one the reconstruction fixes

856 Data and code availability

857 The Lean development, simulation scripts, tests and saved numerical sum-
858 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
859 `cortical_coherence`. The formalization uses Lean 4.34.0-rc2 and Math-
860 lib commit `5ce203ecd53f8c3e47ee5530d7c86b94a55f0183`. Publication
861 simulation values are generated from saved summaries without rerunning
862 production sweeps. The exploratory EEG analysis uses OpenNeuro dataset
863 `ds005620` (Bajwa et al., 2025) (subjects 1010, 1016, 1022, 1033, 1045,
864 1054, 1060, 1067, resting awake and sedation runs 1–2); preprocessing
865 and selection are described in Supplemental Material and implemented in

866 `simulations/empirical_collapse.py`. The repository contains the repro-
867 ducible analysis and does not redistribute the raw recordings.

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871 **Use of AI tools**

872 AI assistance to this work was substantial and is not separable by task.
873 Several coding assistants and chat interfaces were used in an interleaved way
874 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
875 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra and Sol; GPT-
876 5.6 Terra, Luna and Sol), Claude Code (Claude Opus 5 and Opus 5.5),
877 Gemini Pro through its web interface, and Anthropic’s Fable and DeepSeek
878 v4 Flash through OpenRouter, accessed with the Hermes CLI. Between them
879 they contributed to the Lean development, the simulation and analysis code,
880 and the drafting and revision of both documents. Because the tools were
881 used in combination and in alternation, no section, proof, figure or result is
882 attributable to any one of them, and none is claimed to be. The author is
883 responsible for all content, including everything a tool produced. Machine
884 checking establishes the stated Lean results under their hypotheses; it does

885 not validate biological interpretations or replace scientific review of generated
886 code and prose.

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